Summary

Physical methods of microbial control include heat, radiation, drying and filtration.

Heat is the most widely used method of microbial control. It is used in both forms: moist and dry. The thermal death time (TDT) is the time required to kill all microbes at a specific temperature. The thermal death point (TDP) is the lowest temperature at which all microbes are killed in a specified duration of time.

Autoclaving, or steam sterilization, is the process by which steam is heated under pressure to sterilize a wide range of materials in a comparatively short time.

Dry heat kills the microorganisms under specified time and temperature. Dry heat is applied in the following ways: Red heat, incineration and Hot air oven.

Ionizing radiation (cold sterilization) by X rays and gamma rays is used to sterilize medical products and meat. It damages DNA and cell organelles by producing disruptive ions. Ultraviolet light, or nonionizing radiation, has limited penetrating ability. It is therefore restricted to sterilize suface of the materials.

Decontamination by filtration removes microbes from heat sensitive liquids and circulating air. The pore size of the filter determines what kinds of microbes are removed.

Evaluation

Multiple choice questions

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1. Which of the following does not kill endospores?
   a. Autoclaving
   b. Incineration
   c. Hot-air sterilization
   d. Pasteurization
2. Which of the following is most effective for sterilizing Petri dishes?
   a. Chlorine
   b. Ethylene oxide
   c. Autoclaving
   d. Nonionizing radiation
3. _____ kills organisms by coagulation and denaturing their proteins
   a. Dry heat
   b. Moist heat
   c. Both a & b
   d. None of the above
4. In which method, temperature of 160°C for 1 hour is employed?
   a. Red heat
   b. Infrared radiation
   c. Hot air oven
   d. Flaming
5. Which of the following temperature and time are employed in autoclave for sterilization of materials?
   a. 16 lbs 120°C for 18 min
   b. 18 lbs 180°C for 20 min
   c. 22 lbs 170°C for 35 min
   d. 15 lbs 121°C for 15 min
6. Wavelength used for the absorption of UV spectrum is
   a. 4000A°
   b. 2600A°
   c. 20A°
   d. None of the above

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